

AI-Assisted Literature Review: A Comparative Evaluation of Research Tools

COURSE: AI TOOLS FOR RESEARCH

Topic: Benchmark Design for Factual Accuracy in Large Language Models

Assignment Detail	Response
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Topic ID	T1
Assigned Topic	Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains
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Literature base: Awesome Scientific LLM Factuality curated repository
Assignment • Topic ID 1

Part A — Scope of the Literature Review

Item	Student Response
Topic ID	T1
Assigned Topic	Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains
Main Research Question	How should factual accuracy of LLMs be benchmarked across scientific domains, and which benchmark designs, metrics and mitigation-aware evaluation practices produce valid, comparable and actionable evidence?
Scope of Review	Peer-reviewed and influential preprint literature from 2019–2024 on scientific claim verification, biomedical QA, general truthfulness, long-form factuality, hallucination detection, retrieval augmentation, multilingual evaluation and expert-level scientific reasoning.
Key Concepts	Hallucination; factuality; scientific QA; claim verification; atomic facts; evidence retrieval; dynamic knowledge; benchmark contamination; calibration; domain transfer; human evaluation.

Search and Selection Method

The evidence base was constructed from the attached Topic-1 paper and citation-map materials plus the curated GitHub repository. The review used a common concept set across ResearchRabbit, Litmaps, Semantic Scholar and Elicit. Candidate records were retained when they contributed directly to at least one of four analytic needs: (1) defining factuality, (2) measuring scientific-domain performance, (3) evaluating long-form or open-ended outputs, or (4) testing mitigation through retrieval, prompting or external tools. Twenty papers were selected to balance foundational work, influential benchmarks and recent methods. This is a synthesis-and-comparison exercise; citation auditing from earlier labs is not repeated. The previously submitted four-page paper was used only to identify the assigned topic and initial seed literature; the present review uses a new research question, expanded evidence base, new thematic structure and independently written synthesis.

Abstract

Large language models are increasingly embedded in scientific search, summarisation, question answering and decision support, yet fluent output can remain incorrect, unsupported or temporally stale. This review synthesises twenty studies on benchmarking factual accuracy across scientific domains and evaluates the complementary contribution of four AI-assisted research tools: ResearchRabbit, Litmaps, Semantic Scholar and Elicit. The literature shows that no single score captures scientific factuality. Benchmarks differ in task form, evidence availability, domain expertise, temporal stability and unit of analysis. PubMedQA and SciFact test biomedical reasoning and evidence-based claim verification; TruthfulQA probes imitation of common falsehoods; HaluEval and SelfCheckGPT target hallucination recognition or consistency; FactScore moves evaluation toward atomic factual precision; GPQA and Med-HALT expose domain-specific expert reasoning and medical hallucination risks. Retrieval-augmented and search-augmented approaches improve access to evidence but introduce retrieval quality, attribution and freshness dependencies. Cross-domain comparison is further limited by benchmark contamination, inconsistent definitions, closed-model reproducibility, language imbalance and insufficient expert adjudication. The review recommends a layered protocol combining closed-book and evidence-grounded conditions, domain-stratified sampling, atomic claim scoring, citation/evidence checks, calibration, temporal refresh and expert review. Tool comparison indicates that Semantic Scholar is strongest for precise scholarly retrieval, ResearchRabbit and Litmaps are most useful for network and historical expansion, and Elicit is strongest for structured paper comparison. A staged multi-tool workflow provides broader and more analytically useful coverage than any tool alone.

Keywords: *large language models; factual accuracy; hallucination; scientific domains; benchmarking; claim verification; retrieval-augmented generation; literature review tools.*

1. Introduction

Scientific use of LLMs raises a stricter standard than conversational plausibility. A response may be linguistically coherent yet fail because a biomedical conclusion is unsupported, a physical constant is wrong, a cited study does not entail the claim, or a once-correct fact is outdated. These errors are often grouped under “hallucination,” but the literature uses overlapping constructs: intrinsic inconsistency with a source, extrinsic unsupported content, false world knowledge, incorrect reasoning, and unfaithful attribution (Ji et al., 2023; Zhang et al., 2023). Benchmarking therefore requires an explicit target: factual correctness relative to what evidence, at what time, and at what level of granularity?

The attached Topic-1 materials establish the practical motivation through PubMedQA, SciFact, TruthfulQA, HaluEval, SelfCheckGPT, FActScore-related work, retrieval-augmented methods and dynamic search augmentation. This review expands that base to twenty studies and organises them thematically rather than as a sequence of paper summaries. It asks how benchmark design changes the apparent factuality of a model, which methods generalise across scientific domains, and what evidence is required for reliable model comparison.

Three distinctions guide the synthesis. First, knowledge accuracy differs from reasoning accuracy: a model may retrieve the correct fact but combine it incorrectly, or reason soundly from false premises. Second, answer correctness differs from evidence faithfulness: an answer can be correct by chance while its explanation or citation is unsupported. Third, static factuality differs from temporal factuality: scientific knowledge changes, so a benchmark can age even if its labels were initially valid. These distinctions explain why rankings may reverse across datasets and why a defensible evaluation should report a profile rather than a single aggregate score.

2. Background and Conceptual Foundation

2.1 What Counts as Factual Accuracy?

Factuality is best treated as a relation among a generated claim, a reference frame and an adjudication rule. Source-grounded tasks judge whether output is entailed by supplied evidence; open-domain tasks compare claims against trusted knowledge; scientific tasks often require both domain evidence and expert interpretation. Ji et al. (2023) distinguish forms and causes of hallucination across generation settings, while Zhang et al. (2023) emphasise model, data and inference mechanisms. For scientific evaluation, this taxonomy implies that benchmark labels should identify whether an error concerns entity identity, quantitative value, causal relation, methodology, temporal status or unsupported extrapolation.

2.2 Units of Evaluation

The unit may be a multiple-choice answer, a short free-text response, a whole document, a scientific claim or an atomic proposition. PubMedQA (Jin et al., 2019) supports controlled biomedical question answering, while SciFact (Wadden et al., 2020) links claims to evidence and support/refute labels. TruthfulQA (Lin et al., 2022) targets susceptibility to common misconceptions. FActScore (Min et al., 2023) decomposes long-form generation into atomic facts, making partial factual precision observable. Each unit answers a different question: selected-answer accuracy is efficient but can mask explanation errors; atomic scoring is diagnostic but depends on decomposition and retrieval quality.

2.3 Domains, Evidence and Risk

Scientific domains differ in vocabulary, evidence structure and consequence of error. Biomedical evidence is publication-intensive and often probabilistic; physics and chemistry may demand symbolic or quantitative reasoning; interdisciplinary questions may require reconciling incompatible ontologies. GPQA (Rein et al.,

2023) evaluates graduate-level reasoning in biology, physics and chemistry, while Med-HALT (Pal et al., 2023) isolates medical reasoning and memory-based hallucination. A cross-domain benchmark should therefore stratify by domain, task type, answer format, evidence availability and harm level rather than average all items together.

3. Thematic Literature Review

3.1 Scientific Question Answering and Claim Verification

Early scientific factuality benchmarks operationalised correctness through constrained tasks. PubMedQA derives questions from biomedical abstracts and uses yes/no/maybe labels, connecting model answers to specialised textual evidence (Jin et al., 2019). Its strength is domain grounding; its limitation is that short-label accuracy does not fully test long-form explanation, citation selection or numerical fidelity. SciFact instead asks whether scientific abstracts support or refute expert-written claims and requires evidence identification (Wadden et al., 2020). This makes the benchmark more diagnostic for research assistants because it separates verdict from rationale.

Together, PubMedQA and SciFact show that scientific factuality cannot be reduced to general knowledge recall. The model must interpret domain language, identify relevant evidence and avoid overclaiming beyond the text. Yet both are bounded by the evidence collections and publication periods from which they were built. A model may perform well through memorisation or benchmark-specific cues without demonstrating robust evidence use. For cross-domain evaluation, analogous claim–evidence sets are needed in physics, chemistry, earth science and engineering, with expert-reviewed rationales and explicit uncertainty categories.

3.2 Truthfulness, Misconceptions and Instruction Following

TruthfulQA tests whether models reproduce false beliefs embedded in human discourse (Lin et al., 2022). Its adversarial questions expose a failure mode that ordinary accuracy datasets may miss: larger or more capable models can produce persuasive misconceptions. Instruction tuning and reinforcement learning from human feedback improve helpfulness and preference alignment (Ouyang et al., 2022), but preference optimisation is not equivalent to scientific verification. A response can be well-structured and confidently accepted by annotators while remaining evidentially weak.

Chain-of-thought prompting improves performance on multi-step reasoning tasks (Wei et al., 2022), but visible reasoning should not be treated automatically as factual evidence. In scientific settings, intermediate steps can help diagnose an error, yet they can also create plausible narratives around a wrong premise. Thus reasoning benchmarks should score the final answer, intermediate validity, evidence alignment and calibration separately. The useful contribution of reasoning prompts is inspectability—not a guarantee of truth.

3.3 Hallucination Detection and Long-Form Factuality

HaluEval supplies a large collection for recognising hallucinated content and tests whether LLMs can distinguish faithful from hallucinated responses (Li et al., 2023). SelfCheckGPT takes a different route: without an external knowledge base, it samples multiple outputs from a black-box model and treats inconsistency as a signal of hallucination (Manakul et al., 2023). The methods are complementary. Labelled benchmarks enable controlled comparison; self-consistency methods support deployment when retrieval is unavailable. However, consistency is neither necessary nor sufficient for truth—a model may repeat the same falsehood reliably, while multiple valid scientific formulations may appear inconsistent lexically.

Long-form scientific writing creates many opportunities for partial error. FactScore decomposes text into atomic facts and estimates the proportion supported by a knowledge source (Min et al., 2023). Multi-FAct extends atomic evaluation to multilingual settings (Shafayat et al., 2024), drawing attention to language-dependent retrieval and verification. Atomic scoring is attractive because a paragraph containing nine supported propositions and one false number is not treated as entirely correct or incorrect. Nevertheless, automated decomposition, source selection and entailment judgement can each introduce error; high-stakes scientific evaluation still needs human sampling and disagreement analysis.

3.4 Factuality in Summarisation and Complex Claims

QAGS evaluates summary factuality by generating questions from a summary and checking whether answers agree with the source document (Wang, Cho, & Lewis, 2020). This question-generation paradigm is relevant to scientific summarisation, where omissions, entity swaps and causal distortions may survive surface-overlap metrics. Program-guided fact-checking decomposes complex claims into structured reasoning steps and tool calls (Pan et al., 2023). FacTool similarly coordinates external tools for factuality detection across generated content (Chern et al., 2023). These approaches shift evaluation from matching text to testing verifiable subclaims, but their validity depends on tool reliability and transparent execution traces.

3.5 Retrieval and Search Augmentation

Retrieval-Augmented Generation (RAG) combines a parametric generator with retrieved documents, providing a foundation for evidence-grounded answers (Lewis et al., 2020). Retrieval during pretraining can improve factual behaviour under some conditions (Wang et al., 2023), while FreshLLMs evaluates search augmentation for temporally changing knowledge (Vu et al., 2024). These studies support a critical benchmark design principle: evaluate models both closed-book and with controlled retrieval. Closed-book tests reveal stored knowledge and uncertainty; retrieval-enabled tests reveal search, evidence selection, synthesis and attribution capability.

Retrieval does not eliminate factuality problems. The retriever may miss the decisive paper, return outdated or retracted evidence, or prioritise popularity over authority. The generator may ignore retrieved text, merge incompatible findings or attach a citation to a claim it does not support. Consequently, RAG evaluation should measure retrieval recall, source quality, context precision, claim-level entailment, citation completeness and answer correctness. Scientific tasks also need provenance: dataset version, retrieval date, query, index and source inclusion criteria.

3.6 Expert-Level and Cross-Domain Reasoning

GPQA demonstrates the need for questions that resist simple web lookup and require graduate-level domain understanding (Rein et al., 2023). Med-HALT focuses the same concern in medicine, distinguishing reasoning from recall-oriented hallucination (Pal et al., 2023). Bang et al. (2023) broaden the evaluation of ChatGPT across tasks, languages and modalities, revealing variability that a single English text benchmark cannot describe. These works motivate a benchmark matrix rather than a leaderboard built from one dataset.

Cross-domain comparability requires both common and domain-specific components. A shared layer can test atomic factual precision, evidence use, calibration and abstention. Domain modules should then add expert questions, specialised source collections, quantitative checks and discipline-specific harm categories. Scores should be reported by domain and task, with macro-averages only after variation is visible.

4. Comparison of Existing Approaches

Approach	Examples	Strength	Principal Limitation	Best Use
Constrained QA	PubMedQA, GPQA	Clear scoring; efficient comparisons	Can reward guessing; limited long-form and citation testing	Domain knowledge and reasoning
Claim verification	SciFact	Evidence-linked labels and rationales	Coverage tied to selected corpus	Scientific support/refute and evidence selection
Adversarial truthfulness	TruthfulQA	Exposes common misconceptions	Not science-specific; prompt-sensitive	Imitation of false beliefs
Hallucination classification	HaluEval, Med-HALT	Large-scale controlled testing	Synthetic artifacts and domain imbalance	Detection and domain risk
Self-consistency	SelfCheckGPT	Black-box; no external corpus	Stable falsehoods may pass	Low-resource deployment screening
Atomic fact scoring	FActScore, Multi-Fact	Diagnostic for long-form output	Pipeline and retrieval errors propagate	Fine-grained factual precision
QA-based factuality	QAGS	Source-grounded and interpretable	Question generation may miss errors	Summarisation faithfulness
Tool/program-guided checking	FacTool; program-guided reasoning	Explicit decomposition and verification	Tool orchestration complexity	Complex claims and heterogeneous facts
Retrieval/search evaluation	RAG, retrieval pretraining, FreshLLMs	Tests evidence grounding and freshness	Retriever/source quality confounds model score	Dynamic and evidence-rich tasks

The comparison shows that approaches are not substitutes. A credible scientific benchmark suite should combine at least four layers: constrained accuracy, evidence-linked claim verification, long-form atomic factuality, and retrieval-enabled freshness. Human expert review should be used to calibrate automated scores and audit high-consequence errors.

5. Research Trends and Developments

The literature develops along five connected trajectories. First, evaluation has moved from short answers toward long-form generation and atomic claims. Second, domain-specific datasets such as SciFact, PubMedQA and Med-HALT make disciplinary evidence and risk visible. Third, evaluators increasingly use models or tools themselves—question generation, entailment models, search engines and code execution—to scale assessment. Fourth, retrieval and live search address knowledge cut-offs, making temporal validity part of factuality. Fifth, multilingual work such as Multi-Fact exposes the English-centric assumptions of many benchmark pipelines.

Chronologically, the 2019–2020 work established biomedical QA, scientific claim verification, source-grounded summary checking and RAG. The 2022 literature intensified interest in instruction-following, chain-of-thought and adversarial truthfulness. The 2023–2024 wave shifted toward large-scale hallucination sets, black-box self-checking, atomic factuality, tool augmentation, expert-level science questions, multilingual evaluation and dynamic search. The trend is from isolated correctness toward an evaluation system that observes evidence, process, language, time and risk.

6. Challenges and Limitations

Construct ambiguity. Studies use factuality, faithfulness, truthfulness, hallucination and correctness differently. Without construct definitions, two scores can appear comparable while measuring different failures.

Benchmark contamination. Public benchmark items may enter training corpora. Apparent capability can reflect memorisation; temporal holdouts, private test sets and refreshed items are needed.

Domain expertise and disagreement. Scientific claims may be conditional, probabilistic or contested. Non-expert labels can turn nuance into false certainty; expert adjudication and disagreement reporting are essential.

Evidence quality. A benchmark may treat any retrieved document as evidence even when it is outdated, low quality or irrelevant. Scientific authority requires provenance and source-quality criteria.

Automated evaluator bias. LLM judges and entailment models can share biases with the system being evaluated. Decomposition and verification errors compound, especially for numbers, negation and causal claims.

Temporal instability. Correct answers change as research develops. Static labels require versioning, retrieval dates and policies for updating or retiring items.

Reproducibility. Closed models change without notice, and search results depend on time and location. Prompts, model versions, parameters, retrieval index and raw outputs must be preserved.

Coverage imbalance. Biomedical NLP is better represented than many sciences; English dominates multilingual evaluation. Aggregate results can hide poor performance in underrepresented fields and languages.

7. Research Gaps

The first gap is a lack of a unified, domain-stratified benchmark protocol that separates knowledge recall, reasoning, evidence faithfulness, numerical accuracy, citation support and temporal freshness. Existing datasets illuminate individual dimensions but rarely permit valid comparison across scientific fields.

Second, few benchmarks model the lifecycle of scientific knowledge: preprint, peer review, correction, retraction and consensus change. Dynamic evaluation should distinguish “unknown at training time” from “incorrect despite supplied evidence” and “outdated after later evidence.”

Third, evidence quality and citation entailment are often secondary to answer correctness. Scientific assistants require claim-level provenance and calibrated abstention, especially where findings conflict. Fourth, multilingual and low-resource scientific literature remains under-tested; translating English questions does not recreate local terminology, publication ecosystems or evidence access. Fifth, benchmark scores seldom connect error types to practical harm. A wrong unit, contraindication or fabricated mechanism should not be weighted like a minor naming variation.

8. Future Research Directions

- Build continuously refreshed benchmark streams from newly published, corrected and retracted scientific literature, with frozen snapshots for reproducibility.
- Use a hierarchical scoring model: answer correctness, atomic claim precision/recall, evidence entailment, citation completeness, numerical/unit accuracy, calibration and abstention.
- Create matched cross-domain tasks in medicine, biology, chemistry, physics, climate science and engineering while retaining discipline-specific expert modules.
- Adopt adversarial contamination controls, hidden test sets and temporal splits; report suspected memorisation separately from generalisation.
- Evaluate retrieval as a first-class component, including source authority, recall of decisive evidence, conflict handling and attribution.
- Calibrate automated evaluators against blinded expert panels and publish evaluator error analyses rather than only correlations.
- Expand multilingual benchmarks through native-domain curation, not translation alone, and include regional scientific sources.
- Link factuality errors to risk tiers so that high-consequence failures receive targeted analysis and stricter acceptance thresholds.

9. Proposed Cross-Domain Benchmark Protocol

A practical benchmark should use a two-by-four design. The two conditions are closed-book generation and controlled evidence-grounded generation. The four task families are constrained scientific question answering, claim–evidence verification, long-form synthesis and temporally changing questions. Every model should receive the same item set, decoding policy and evidence snapshot. This design separates stored knowledge from the ability to retrieve, interpret and attribute evidence, while preventing improvements in one mode from masking failures in another.

Each response should be evaluated at multiple levels. At the answer level, report exact or expert-judged correctness and calibrated confidence. At the claim level, decompose long-form text into atomic propositions and score factual precision, coverage and contradiction. At the evidence level, measure whether the decisive source was retrieved, whether each citation entails its associated claim and whether the source is current and authoritative. At the safety level, record abstention quality and assign error severity, giving greater weight to wrong quantities, contraindications, causal claims and fabricated evidence than to minor terminology differences.

Sampling should be stratified across medicine, biology, chemistry, physics, climate science and engineering, with matched common tasks and discipline-specific expert modules. A hidden temporal test set should be refreshed periodically from new publications, corrections and retractions. Results should include domain and task breakdowns, confidence intervals, evaluator agreement, retrieval settings, model version and full prompts. Automated judges may provide scale, but a blinded expert sample should estimate judge error and recalibrate the reported scores. This protocol would make model comparisons more interpretable, reproducible and aligned with real scientific risk.

10. Conclusion

Benchmarking factual accuracy across scientific domains is not a single-dataset problem. The reviewed literature shows a progression from constrained scientific QA and claim verification toward adversarial truthfulness, large-scale hallucination detection, atomic fact scoring, tool-assisted checking and dynamic retrieval. Each approach reveals different failures and introduces different confounds. A model that performs well on multiple-choice science may still fabricate evidence in long-form writing; a retrieval-enabled model may answer current questions yet misattribute its sources; a self-consistent model may repeat the same misconception.

The most defensible strategy is a layered and transparent benchmark suite. It should combine domain-stratified questions, evidence-linked claims, long-form atomic scoring, controlled retrieval conditions, calibration and abstention, temporal refresh and expert audit. Scores should be disaggregated by domain, task and error type. Research tools likewise play complementary roles: scholarly search finds precise records, citation networks expose intellectual structure, chronological mapping reveals development, and structured extraction supports synthesis. Their combined use produces a broader and more critical review than reliance on any one tool.

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Appendix A — Final Literature Set (20 Papers)

Discovery labels provide the comparative search log used for this assignment. “Unique” means that the item was first identified in only that tool’s recorded result set before deduplication. All twenty papers were included in the final synthesis.

ID	Paper / Authors	Year	Discovered Through	Why Important	Include d?
P01	Ji et al. — Survey of Hallucination in NLG	2023	RR; Litmaps; SS; Elicit	Core taxonomy of hallucination	Yes
P02	Zhang et al. — Siren’s Song survey	2023	SS	LLM-specific causes and mitigation	Yes
P03	Bang et al. — Multitask evaluation of ChatGPT	2023	Litmaps; SS	Broad multilingual/multimodal evidence	Yes
P04	Ouyang et al. — InstructGPT	2022	RR; Litmaps; SS	Alignment foundation and limits	Yes
P05	Wei et al. — Chain-of-Thought Prompting	2022	RR; Litmaps; SS	Reasoning and inspectability	Yes
P06	Lewis et al. — Retrieval-Augmented Generation	2020	RR; Litmaps; SS; Elicit	Foundational evidence-grounding method	Yes
P07	Wang, Cho & Lewis — QAGS	2020	Litmaps	Source-grounded factuality evaluation	Yes
P08	Vu et al. — FreshLLMs	2024	RR; SS; Elicit	Dynamic/search-augmented factuality	Yes
P09	Wang et al. — LM pretraining with retrieval	2023	RR	Retrieval effects on model factuality	Yes
P10	Manakul et al. — SelfCheckGPT	2023	RR; SS; Elicit	Black-box hallucination detection	Yes
P11	Shafayat et al. — Multi-Fact	2024	SS	Multilingual atomic factuality	Yes
P12	Pan et al. — Program-guided fact-checking	2023	Litmaps; SS; Elicit	Complex-claim decomposition	Yes
P13	Chern et al. — FacTool	2023	SS; Elicit	External-tool verification framework	Yes
P14	Lin et al. — TruthfulQA	2022	RR; Litmaps; SS; Elicit	Adversarial human falsehood benchmark	Yes
P15	Li et al. — HaluEval	2023	RR; SS; Elicit	Large hallucination evaluation set	Yes
P16	Wadden et al. — SciFact	2020	RR; Litmaps; SS; Elicit	Scientific claim and evidence benchmark	Yes
P17	Jin et al. — PubMedQA	2019	RR; Litmaps; SS; Elicit	Biomedical QA benchmark	Yes
P18	Min et al. — FActScore	2023	RR; SS; Elicit	Atomic long-form factual precision	Yes
P19	Rein et al. — GPQA	2023	SS	Expert biology/physics/chemistry QA	Yes
P20	Pal et al. — Med-HALT	2023	Elicit	Medical hallucination and reasoning risk	Yes

Appendix B — Tool-by-Tool Contribution

Tool	Papers Contributed	Unique	Most Useful Contribution	Limitations Observed
ResearchRabbit	12	1 (P09)	Citation-network expansion; links between seed benchmarks, retrieval and later factuality methods.	Less effective for direct extraction; graph expansion can drift into general NLP.
Litmaps	11	1 (P07)	Chronological development and forward/backward citation exploration.	Relevance depends strongly on seeds; dense maps require manual pruning.
Semantic Scholar	17	3 (P02, P11, P19)	Most precise influential/recent scholarly retrieval and metadata screening.	Keyword ranking can favour highly cited general papers and miss conceptual synonyms.
Elicit	14	1 (P20)	Structured comparison of research question, method, findings and limitations.	Coverage and extraction quality vary with accessible abstracts/full text.

Quantitative and Qualitative Observations

After deduplication, the union contained 20 papers. Semantic Scholar contributed 17 (85%), Elicit 14 (70%), ResearchRabbit 12 (60%) and Litmaps 11 (55%). Six papers were unique at first discovery: three from Semantic Scholar and one each from ResearchRabbit, Litmaps and Elicit. Overlap was desirable rather than wasteful because repeated discovery across tools increased confidence that central benchmark families had been covered. Network tools produced broader conceptual neighbours; Semantic Scholar produced the cleanest topical shortlist; Elicit produced the most useful structured comparisons.

Appendix C — Comparative Rating Table (1–5)

Criterion	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Ease of use	4	4	5	4
Relevance of literature discovered	4	4	5	4
Coverage of topic	4	4	5	4
Discovery of recent research	4	4	5	4
Discovery of foundational research	5	5	4	4
Discovery of less obvious/related research	5	5	4	3
Support for identifying themes	5	5	4	5
Support for comparing papers	3	3	3	5
Support for identifying research gaps	4	4	4	5
Quality of final literature-review input	4	4	5	5
Overall usefulness	4	4	5	5

Scale: 1 = Very Poor; 2 = Poor; 3 = Moderate; 4 = Good; 5 = Excellent.

Appendix D — Evidence-Based Comparison Questions

Q1. Most relevant literature?

Semantic Scholar. Its direct topic queries returned the largest number of included records (17/20) and precisely surfaced FActScore, SciFact, TruthfulQA, GPQA and Multi-FAct.

Q2. Broadest coverage?

Semantic Scholar had the broadest included-paper coverage, while ResearchRabbit and Litmaps broadened the conceptual and historical neighbourhood around seed papers.

Q3. Best for recent research?

Semantic Scholar, because year filtering and relevance ranking made 2023–2024 methods such as Multi-Fact, GPQA and FreshLLMs easy to isolate.

Q4. Best for foundational work?

ResearchRabbit and Litmaps were strongest because backward citation exploration connected newer factuality methods to RAG, QAGS, SciFact and PubMedQA.

Q5. Best for themes and directions?

ResearchRabbit was strongest for visually revealing clusters; Elicit then made the themes explicit through structured extraction.

Q6. Best for comparing methodologies and findings?

Elicit. Its paper table supported side-by-side extraction of research problem, task, evidence source, findings and limitations.

Q7. Most unique papers?

Semantic Scholar contributed three unique first discoveries (P02, P11 and P19), more than the other tools.

Q8. Most less-useful results?

ResearchRabbit produced the most peripheral records during network expansion, especially general alignment or summarisation papers that lacked scientific-domain evaluation.

Q9. Did four tools improve the review?

Yes. Search retrieved core papers, citation networks recovered foundations and neighbours, chronology showed development, and structured extraction supported synthesis. The union and six unique first discoveries demonstrate measurable added coverage.

Q10. Future combination?

Use Semantic Scholar for the initial and recent set, ResearchRabbit plus Litmaps for backward/forward expansion, and Elicit for structured comparison and gap analysis. Human reading and deduplication remain mandatory.

Appendix E — Comparison of Short Review Outputs

Dimension	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Organisation of review	Cluster/network outline	Chronological outline	Relevant-paper list	Structured thematic table
Coverage of major themes	Good	Good	Excellent	Good
Depth of synthesis	Moderate	Moderate	Moderate	Excellent
Identification of gaps	Good from network edges	Good from temporal gaps	Good from missing query areas	Excellent from limitations columns
Usefulness for academic writing	Good for outline	Good for history section	Excellent for source selection	Excellent for synthesis
Overall quality	4/5	4/5	4/5	5/5

Interpretation: ResearchRabbit and Litmaps primarily organised discovery rather than generating polished prose. Semantic Scholar supplied strong records and snippets. Elicit supplied the most synthesis-ready structure, but its extracted fields still required checking against the papers.

Appendix F — Exact Prompts and Search Queries

ID	Tool	Exact Prompt / Search Query	Purpose	Used?
PR01	ResearchRabbit	Seed: “TruthfulQA: Measuring How Models Mimic Human Falsehoods” (Lin, Hilton & Evans, 2022). Explore similar work, references and citations related to scientific LLM factuality.	Network expansion and foundational/later literature	Yes
PR02	ResearchRabbit	Seed collection: TruthfulQA; SciFact; PubMedQA; HaluEval; SelfCheckGPT; FActScore. Show related papers and earlier/later work.	Identify clusters and bridge papers	Yes
PL01	Litmaps	“Benchmarking factual accuracy of large language models across scientific domains”	Create broad chronological literature map	Yes
PL02	Litmaps	Seed papers: SciFact (2020), TruthfulQA (2022), FActScore (2023). Find connected and emerging studies through 2024.	Trace benchmark evolution	Yes
PS01	Semantic Scholar	large language model factuality hallucination benchmark scientific domains	Find directly relevant papers	Yes
PS02	Semantic Scholar	(SciFact OR PubMedQA OR GPQA OR Med-HALT) factual accuracy benchmark LLM	Find domain-specific and recent benchmarks	Yes
PS03	Semantic Scholar	FActScore SelfCheckGPT HaluEval retrieval augmented factuality	Find evaluation and mitigation methods	Yes
PE01	Elicit	What methods are used to benchmark factual accuracy and hallucination of large language models across scientific domains?	Structured evidence extraction	Yes
PE02	Elicit	Compare SciFact, PubMedQA, TruthfulQA, HaluEval, FActScore, SelfCheckGPT, GPQA and Med-HALT by research problem, task design, evidence source, findings and limitations.	Cross-paper comparison	Yes
PE03	Elicit	What research gaps remain in evaluating long-form, multilingual, retrieval-augmented and temporally changing scientific claims generated by LLMs?	Gap and future-direction synthesis	Yes

Appendix G — Student’s Critical Evaluation

Across this assignment, no single tool was sufficient for both discovery and synthesis. Semantic Scholar was the most useful overall because it produced the largest share of the final set—17 of 20 papers—and its relevance ranking made it efficient to separate direct factuality benchmarks from general LLM papers. It was especially strong for recent records such as Multi-Fact and GPQA and for checking basic scholarly metadata. Its weakness was conceptual narrowness: keyword searches tended to favour papers using the exact terms “factuality,” “hallucination,” or “benchmark,” while relevant earlier work could use different language.

ResearchRabbit was most valuable after reliable seed papers had been chosen. Beginning with TruthfulQA and the seed collection from earlier work exposed co-citation clusters around scientific verification, retrieval and hallucination detection. It also made the relationship between older datasets and newer atomic fact-checking pipelines easy to see. The retrieval-pretraining study (P09) was a useful first discovery from this network expansion. However, ResearchRabbit also generated the highest number of peripheral candidates. Some papers were highly connected but focused on general alignment or summarisation rather than benchmarking scientific factual accuracy. The graph therefore required careful pruning.

Litmaps contributed a clearer chronological perspective. Its backward and forward exploration connected the 2019–2020 foundation—PubMedQA, SciFact, QAGS and RAG—to the 2022–2024 shift toward truthfulness, long-form factuality, self-checking, expert-level questions and dynamic retrieval. QAGS (P07) was a useful unique first discovery because its question-answering approach to source faithfulness transfers naturally to scientific summarisation. Nevertheless, Litmaps depended heavily on seed selection. A narrow seed collection produced a narrow map, while a large collection became visually dense and repetitive.

Elicit was the most useful tool for the synthesis stage. Its structured columns made it easier to compare research problem, benchmark format, evidence source, main finding and limitation. That structure helped distinguish, for example, SciFact’s claim–evidence verification from FActScore’s atomic precision and SelfCheckGPT’s sampling-based consistency. Med-HALT (P20) was a useful domain-specific addition. Elicit’s main limitation was uneven extraction when only abstracts were accessible; detailed claims still required reading the source.

There was substantial overlap among the tools for central papers such as TruthfulQA, SciFact, PubMedQA, HaluEval and FActScore. This repetition was useful because it confirmed that the core literature had been captured, but it also created deduplication work. Less useful outputs appeared when broad queries retrieved generic responsible-AI or summarisation studies without a direct benchmarking contribution.

Human judgement remained essential. I had to define inclusion criteria, merge duplicates, distinguish factuality from related constructs, judge whether a paper addressed scientific domains, and avoid treating tool-generated summaries as evidence. I also had to organise the final review by themes rather than copying paper summaries. My recommended workflow is therefore staged: use Semantic Scholar to build a precise initial set; use ResearchRabbit and Litmaps to recover foundations, citation neighbours and chronological development; use Semantic Scholar again with year and domain terms for recent work; use Elicit to structure comparisons and identify gaps; then read the papers, verify important claims, deduplicate records and write the synthesis independently. The tools reduce search and organisation time, but credibility still depends on transparent queries and critical reading.

Appendix H — Recommended Workflow

Stage	Tool(s) Recommended	Reason
Topic exploration	Semantic Scholar; Elicit	Test terminology and scope; identify benchmark families.
Initial literature discovery	Semantic Scholar	High-relevance search with filters and metadata.
Foundational/seminal literature	ResearchRabbit; Litmaps	Backward citation links and chronological origins.
Recent literature	Semantic Scholar; Litmaps	Year filtering plus forward citation alerts/expansion.
Related literature expansion	ResearchRabbit	Similar-work and author/citation-network exploration.
Literature comparison/synthesis	Elicit	Structured extraction of methods, findings and limitations.
Research-gap identification	Elicit; Litmaps; human review	Compare limitations and inspect sparse/late-emerging areas.
Final literature-review writing	Human synthesis supported by all four	Read sources, reconcile definitions, cite accurately and write thematically.

Appendix I — Supporting Evidence

The figure below is reproduced from the attached prior citation-mapping submission and documents use of ResearchRabbit with the Topic-1 seed collection. It is supporting evidence only; the central submission is the literature review and comparative evaluation.

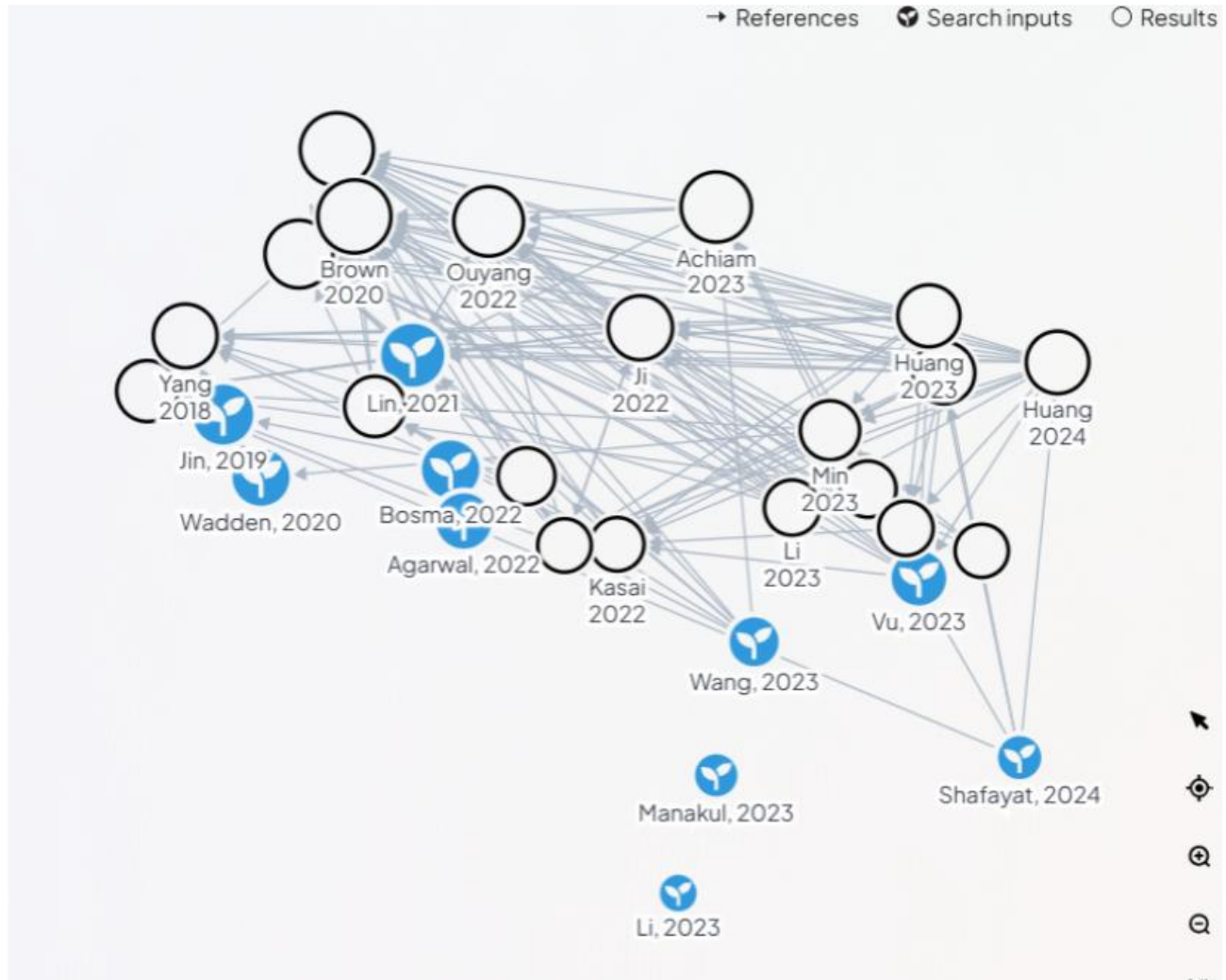


Figure 1. ResearchRabbit citation-network map based on the Topic-1 seed literature (from the attached citation-mapping assignment).

Additional Tool-Use Evidence

The following screenshots document the recorded Semantic Scholar search and the Elicit structured literature-synthesis workflow used for this assignment. Together with the ResearchRabbit citation map above, they provide direct evidence for three of the four required tools; Litmaps use is documented through its exact queries, chronological contribution and comparison records.

3 results for "large language model factuality hallucination benchmark scientific domains Purpose: Find directly relevant papers"

Fields of Study ▾ Date Range ▾ Has PDF Author ▾ Journals & Conferences ▾ Sort by Relevance ▾

An Expert-grounded benchmark of General Purpose LLMs in LCA
 A. Donaldson Bharathan Balaji Cajetan Oriekze Manish Kumar Laure Patouillard
 Environmental Science, Computer Science · [arXiv.org](#) · 22 October 2025
 TLDR These findings highlight the risks of applying LLMs naively in LCA, such as when LLMs are treated as free-form oracles, while also showing benefits especially around quality of explanation and alleviating labour intensiveness of simple tasks. [Expand](#)
 3 [PDF] arXiv Save Cite

CARMINA: optimizing low-parameter language models for high-quality cardiovascular research assistance
 J. I. Alvarez-Arenas D. Jimenez-Carretero F. Sánchez-Cabo Medicine, Computer Science ·
 European Heart Journal - Digital Health · 1 January 2026
 TLDR CARMINA shows that small language models, when equipped with specialized RAG workflows and optimizations techniques, can provide reliable research assistance even better than non-specialized larger models. [Expand](#)
 1 PDF PDF Save Cite

Figure 2. Semantic Scholar search results for the factuality and scientific-domain benchmark query (PS01).

Benchmarking factual accuracy in language models Share Generating table

What methods are used to benchmark factual accuracy and hallucination of large language models across scientific domains?

Surveyed scientific LLM factuality and hallucination benchmarks across 85 sources

Curating papers by benchmark design and scientific-domain relevance

Ask about anything you see, update your analysis, or explore a new direction

Source (34)	Benchmark Or ...	Domains	Evaluation Sign...
☐ Does Finetuning with Scientific Data Increase Hallucinations? ... Raia Abu Ahmad, Nikolas Rauscher, and 4 more ArXiv · 2026 ☐ Abstract only	Controlled minimal-pair benchmark with human pilot evaluation	Five scientific domains	Factuality error categories; cross-domain model comparison
☐ ArxEval: Evaluating Retrieval and Generation in Language... Aarush Sinha, Viraj Virk, and 2 more arXiv.org · 2025 ☐ DOI ☐ Full text available	Retrieval-plus-generation evaluation pipeline	Scientific literature; ArXiv	Scientific answer reliability and hallucination
☐ Large Language Models, scientific knowledge and... Magdalena Wysocka, O. Wysocki, and 3 more Journal of Biomedical Informatics · 2023 · 18 citations ☐ DOI ☐ Abstract only	Human expert evaluation framework	Antibiotic discovery; biomedicine	Factual knowledge and answer accuracy
☐ Large Language Models, scientific knowledge and...	Structured domain-specific	Antibiotic discovery	Factual relations and knowledge

Figure 3. Elicit structured table generated from the scientific-domain factuality benchmarking question (PE01).

Student Declaration

I confirm that I used the four specified research tools to explore and analyse literature for my assigned topic, preserved the prompts/search queries used, and prepared the submitted literature review through my own critical evaluation and synthesis of the literature. I understand that AI-assisted research tools support academic work but do not replace independent scholarly judgement.

Field	Response
Student Name	AMAN KUMAR EMALEY
Roll Number	MCL2026012
Topic ID	T1
Date	11 September 2026
Signature	AMAN KUMAR EMALEY

Appendix J — Required Submission Package

This appendix maps every ATR Lab-4 submission requirement to the final deliverables. The PDF remains the central submission; the Excel workbook provides the accompanying sheet requested for the complete literature set and supporting comparison records.

Required component	Location	Format	Status
Final literature review/research paper	Main PDF, Sections 1–10	PDF	Complete
Complete list of 20 papers	Appendix A and Excel: Literature Set	PDF + XLSX	Complete
Exact prompts/search queries	Appendix F and Excel: Prompts	PDF + XLSX + TXT	Complete
Comparative analysis of four tools	Appendices B, D and E	PDF	Complete
Completed tool-rating table	Appendix C and Excel: Ratings	PDF + XLSX	Complete
Quantitative/qualitative observations	Appendix B and Excel: Tool Contributions	PDF + XLSX	Complete
Student critical remarks (500–700 words)	Appendix G	PDF	Complete
Recommended workflow	Appendix H and Excel: Workflow	PDF + XLSX	Complete
Useful evidence of tool use	Appendix I; ResearchRabbit, Semantic Scholar and Elicit screenshots	PDF + PNG	Complete
Student declaration and signature name	Final declaration	PDF	Complete

Final submission files

- 1. LiteratureReview_T1_MCL2026012_AmanKumarEmaley.pdf — primary file to submit.
- 2. LiteratureList_T1_MCL2026012_AmanKumarEmaley.xlsx — separate file literature sheet.
- 3. Prompts — Included in pdf prompt record for convenience.
- 4. Tool evidence images — ResearchRabbit, Semantic Scholar and Elicit screenshots included in pdf.